

Miaomiao Chen

Sr. Machine Learning Engineer, Ph.D.

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SUMMARY

Machine Learning Engineer with 7+ years in MLOps, Large-Scale Data Analysis, and Agentic AI. Expert in building production-grade LLM systems (RAG, Post-train) and scalable Kubernetes microservices for pharmaceutical needs. China National Chess Champion, bringing strategic depth, resilience, and rigorous mathematical thinking to complex engineering challenges.

EDUCATION

M.S. in Computer Science (Machine Learning Track), Georgia Institute of Technology	2024 - present (Remote)
Ph.D. in Chemistry (Computational Modeling), University of Virginia, VA	2019 - 2024
B.S. in Bioscience, ShanghaiTech University, China	2015 - 2019

TECHNICAL SKILLS

Programming & Infra: Python, SQL, Spark, Nextjs ; Docker, Kubernetes, vLLM
Machine Learning: Transformers(GPT, BERT, VIT, HSTU), Agents (LangChain), PyTorch, Reinforcement Learning (RL)
life Science: Proteomics, Genomics, Multi-Omics, Protein Science

EXPERIENCE

- Sr. Research Scientist (Machine Learning Focus)** | *AstraZeneca PLC* | MD Jul 2024 – Present
- Led the development of domain-specialized AI Agents using **LangChain, RAG, and MonetDB**; integrated analytical tools for multi-cohort comparisons and statistical testing, accelerating cancer patient data interpretation by 3x.
 - Developed encoder-transformer based **Foundation Models** on clinical trial data to forecast drug response rates; achieved a **20% accuracy uplift**, directly accelerating the ranking and prioritization of high-potential therapeutic candidates.
 - Architected an unsupervised Transformer-based anomaly detector for high-dimensional instrument data, designing a novel "run-load gap" embedding to capture complex operational drift superior to standard statistical methods.
 - Engineered a full-stack QC analytical platform (FastAPI, Next.js) deployed on **Kubernetes**; integrated into the core pipeline to automate statistical profiling and visualization, enabling users to rapidly assess sample integrity and instrument health.
 - Designed a **Dense Passage Retrieval (DPR)** system by fine-tuning **BGE embeddings** via **contrastive learning with in-batch negatives**; optimized semantic alignment on 7k+ proteomics Q&A pairs to power high-precision RAG queries.
- Graduate Researcher (Computational Modeling)** | *University of Virginia* | VA 2019 – 2024
- Engineered a sparse proteomics imputation pipeline benchmarking **NMF vs. KNN**; validated via masked RMSE to prove NMF's superior latent structure preservation, resolving key limitations in existing software.
 - Developed an end-to-end predictive modeling pipeline (**Random Forest, XGBoost, LSTM**) for high-dimensional drug candidate prioritization, achieving **85.3% accuracy** and a **20% wet-lab validation hit rate**.
 - Performed data analysis workflows and A/B testing frameworks to identify novel inflammation mechanisms; validated computational insights via in-vivo mouse models, resulting in a **first-author publication in PNAS**.

SELECTED PROJECTS

- Krono: Voice-Activated AI News Agent** 🗣️ 🔄 | *Python, LangGraph, vLLM, GRPO* 2025 – Present
- Architected a **multi-agent system** using **LangGraph** and **SenseVoice ASR** for intent detection and tool routing, enabling natural, interruptible voice interactions with persistent conversational context.
 - Post-trained Qwen-1.5B** using **verl (Ray overrides) + GRPO (Group Relative Policy Optimization)** and deployed via **vLLM**, reducing inference costs by **50%** while maintaining high classification accuracy.
 - Designed a real-time threading architecture to orchestrate audio streams and **RAG retrieval**, achieving **<100ms latency** while reducing user information lookup time by **~70%** compared to manual search.
- GPT-2 Reproduction & Optimization** 📝 🛠️ | *PyTorch, Transformers, CUDA* 2024 – Present
- Built and trained a custom **GPT-2 (124M)** style model from scratch (12 layers, 384-dim), integrating **FlashAttention** to significantly accelerate training throughput and optimize GPU memory utilization.
 - Implemented core Transformer operators (e.g., **Softmax, RoPE, Causal Attention**) from first principles following the **Stanford CS336** curriculum, mastering low-level matrix arithmetic (MatMul) and computation graphs.
 - Validated robustness on the **HellaSwag** benchmark (26% accuracy) after 3,800+ training steps.
 - Implemented **QLoRA** for memory-efficient fine-tuning on multilingual datasets, demonstrating expertise in adapting large models on constrained hardware resources.
- Generative Retrieval Video RecSys (HummingbirdRec)** 🎥 🔄 | *PyTorch, HSTU, Transformers* 2024 – Present
- Architected a SOTA **Generative Recommender** based on **HSTU (Hierarchical Sequential Transduction Unit)** for UGC short-video feeds, outperforming strong baselines (SASRec) by **+11% in HR@10 and NDCG@10**.
 - Engineered a custom **RoPE** mechanism injecting **timestamp and duration-aware signals**, capturing fine-grained temporal dynamics to achieve a **15% NDCG uplift**.
 - Designed a **self-adaptive label normalization** strategy based on **watch-ratio** (playtime/duration), reducing implicit feedback noise and driving an additional **~3% gain** on the **KuaiRec** benchmark.